Visual Data Analysis WS 2026
Assignment MW (2026)

Instructor: Prof. Dr. Alexander Wiebel
Hochschule Worms University of Applied Sciences
wiebel@hs-worms.de - https://teaching.awmw.org

All programming of the multi-week (**MW**) assignment will be implemented in Python using VTK.

Problem MW.1 Preliminary Analysis

Consider all following tasks for **both** the train and the car **data sets** provided through the SeaFile links in the Moodle course.

MW.1.1 Basic Statistics

Load the data set in VTK. Print the following statistics to the console:

- Number of vertices
- Overall number of cells
- Number of cells of a certain type: tetrahedra, prisms, pyramids.
- Maximum and minimum of x, y and z coordinates of all vertices (These form the so called bounding box of the data set)
- (x,y,z) coordinates and values of the points with the highest and lowest pressure
- (x,y,z) coordinates and values of the points with the highest and lowest density
- Mean pressure
- Median pressure
- Mean density
- Median density

MW.1.2 Overview Visualization

- Draw the bounding box of the data set using lines.
- Draw spheres at the following positions: minimum pressure, maximum pressure, minimum density, maximum density.

Problem MW.2 Slice Visualization

Integrate the following into your visualization from above.

MW.2.1 Colored Slice

- Create a slice orthogonal to driving direction of the car/train. The slice shall intersect the car/train. The slice shall encode the *pressure* as color using the viridis color map.
- Provide an expressive color bar as legend.

MW.2.2 Contour Slice

- Use the same slice to draw 10 expressive isolines (*contours*) of the *density*.
- For each contour at least one small number should indicate the isovalue of the isoline.

MW.2.3 Features in the Slice

- Detect the positions of the maximum and the minium of the *pressure* and the *density* in the slice.
- Show the positions using small 3D markers (e.g. spheres, boxes, ...).
- Choose expressive shapes and colors, such that one can easily understand what each marker represents.

This PDF will be extended in the coming weeks in order to provide you with more tasks.